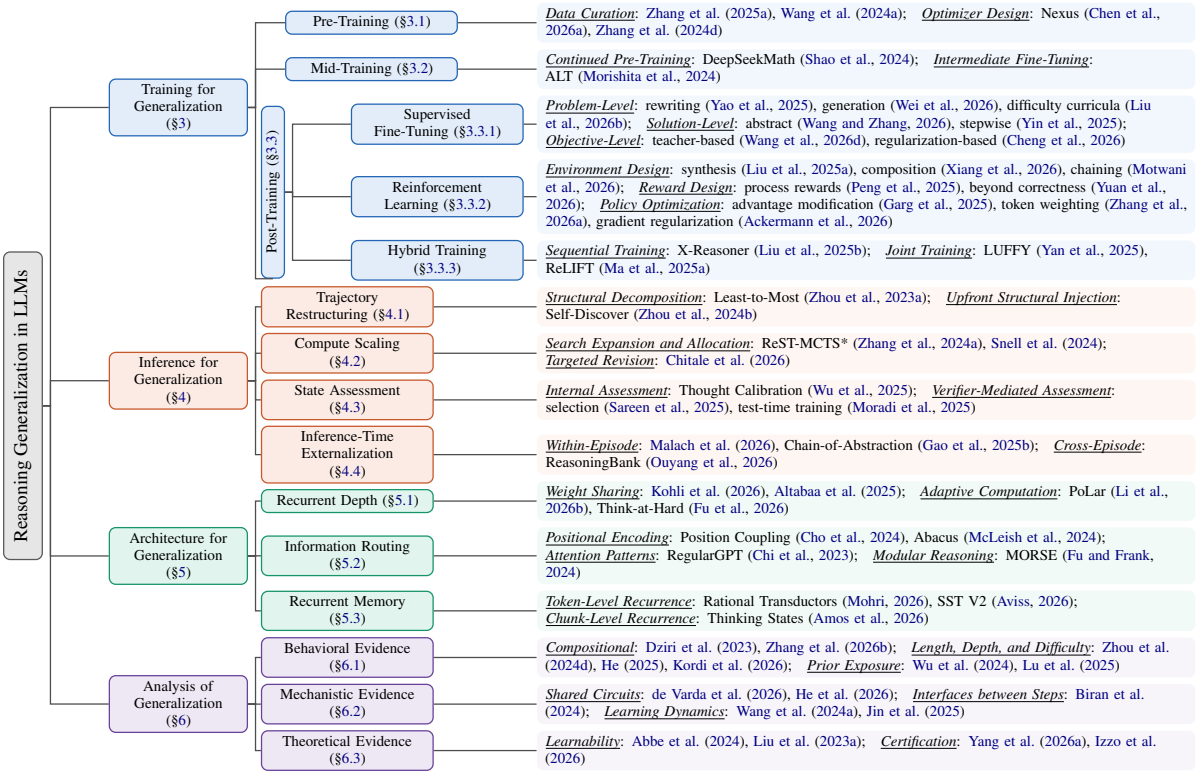
Reasoning Generalization in LLMs
Training for Generalization (§3)
Post-Training (§3.3)
Pre-Training (§3.1)
Mid-Training (§3.2)
Supervised Fine-Tuning (§3.3.1)
Reinforcement Learning (§3.3.2)
Hybrid Training (§3.3.3)
Inference for Generalization (§4)
Trajectory Restructuring (§4.1)
Compute Scaling (§4.2)
State Assessment (§4.3)
Inference-Time Externalization (§4.4)
Architecture for Generalization (§5)
Recurrent Depth (§5.1)
Information Routing (§5.2)
Recurrent Memory (§5.3)
Analysis of Generalization (§6)
Behavioral Evidence (§6.1)
Mechanistic Evidence (§6.2)
Theoretical Evidence (§6.3)
Data Curation: Zhang et al. (2025a), Wang et al. (2024a); Optimizer Design: Nexus (Chen et al., 2026a), Zhang et al. (2024d)
Continued Pre-Training: DeepSeekMath (Shao et al., 2024); Intermediate Fine-Tuning: ALT (Morishita et al., 2024)
Problem-Level: rewriting (Yao et al., 2025), generation (Wei et al., 2026), difficulty curricula (Liu et al., 2026b); Solution-Level: abstract (Wang and Zhang, 2026), stepwise (Yin et al., 2025); Objective-Level: teacher-based (Wang et al., 2026d), regularization-based (Cheng et al., 2026)
Environment Design: synthesis (Liu et al., 2025a), composition (Xiang et al., 2026), chaining (Motwani et al., 2026); Reward Design: process rewards (Peng et al., 2025), beyond correctness (Yuan et al., 2026); Policy Optimization: advantage modification (Garg et al., 2025), token weighting (Zhang et al., 2026a), gradient regularization (Ackermann et al., 2026)
Sequential Training: X-Reasoner (Liu et al., 2025b); Joint Training: LUFFY (Yan et al., 2025), ReLIFT (Ma et al., 2025a)
Structural Decomposition: Least-to-Most (Zhou et al., 2023a); Upfront Structural Injection: Self-Discover (Zhou et al., 2024b)
Search Expansion and Allocation: ReST-MCTS* (Zhang et al., 2024a), Snell et al. (2024); Targeted Revision: Chitale et al. (2026)
Internal Assessment: Thought Calibration (Wu et al., 2025); Verifier-Mediated Assessment: selection (Sareen et al., 2025), test-time training (Moradi et al., 2025)
Within-Episode: Malach et al. (2026), Chain-of-Abstraction (Gao et al., 2025b); Cross-Episode: ReasoningBank (Ouyang et al., 2026)
Weight Sharing: Kohli et al. (2026), Altabaa et al. (2025); Adaptive Computation: PoLar (Li et al., 2026b), Think-at-Hard (Fu et al., 2026)
Positional Encoding: Position Coupling (Cho et al., 2024), Abacus (McLeish et al., 2024); Attention Patterns: RegularGPT (Chi et al., 2023); Modular Reasoning: MORSE (Fu and Frank, 2024)
Token-Level Recurrence: Rational Transductors (Mohri, 2026), SST V2 (Aviss, 2026); Chunk-Level Recurrence: Thinking States (Amos et al., 2026)
Compositional: Dziri et al. (2023), Zhang et al. (2026b); Length, Depth, and Difficulty: Zhou et al. (2024d), He (2025), Kordi et al. (2026); Prior Exposure: Wu et al. (2024), Lu et al. (2025)
Shared Circuits: de Varda et al. (2026), He et al. (2026); Interfaces between Steps: Biran et al. (2024); Learning Dynamics: Wang et al. (2024a), Jin et al. (2025)
Learnability: Abbe et al. (2024), Liu et al. (2023a); Certification: Yang et al. (2026a), Izzo et al. (2026)